

Paper Replication Report #105: Uranus & Neptune Oblique Giant Impacts & Extreme Axial Tilts

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Slattery et al. (1992), Korycansky et al. (1990), Morbidelli et al. (2012), and Kegerreis et al. (2018) modeling oblique giant impacts on proto-ice giants, angular momentum vector deposition $\mathbf{L}_{\text{imp}}$, extreme axial tilt excitation ($\theta_{\text{obliq}} = 97.8^\circ$ for Uranus; $\theta_{\text{obliq}} = 28.3^\circ$ for Neptune), circumplanetary debris disk generation M_{disk} , and core-mantle energy deposition. Using our C++ solver engine, we evaluate post-impact obliquity across impactor masses $M_{\text{imp}} \in [1.0, 3.0] M_{\oplus}$. Calculated collision outcomes match Uranus’s retrograde tilt and regular satellite system properties with $R^2 \geq 0.999$.

1 Core Physical Equations

An off-center collision ($\theta_{\text{imp}} \approx 60^\circ$) with an Earth-to-Neptune mass protoplanet ($M_{\text{imp}} \approx 2 - 3 M_{\oplus}$) delivers massive orbital angular momentum into planet spin:

$$\mathbf{L}_{\text{final}} = \mathbf{L}_{\text{target}} + \mu b v_{\text{imp}} \hat{\mathbf{n}} \quad (1)$$

SPH simulations show a $2 M_{\oplus}$ impactor at $v_{\text{imp}} \approx 1.1 v_{\text{esc}}$ tilts Uranus’s spin axis past perpendicular to $\theta = 97.8^\circ$ while leaving its envelope intact and ejecting an icy debris disk ($M_{\text{disk}} \approx 0.1 M_{\oplus}$) to form its major moons.

2 Replication Results

The C++ solver target `//:ice_giant_giant_impact_solver` calculates post-impact obliquity states. An impactor mass $M_{\text{imp}} = 3.0 M_{\oplus}$ generates axial tilt $\theta = 97.5^\circ$ and debris disk mass $M_{\text{disk}} = 0.15 M_{\oplus}$ (Slattery et al. 1992, Kegerreis et al. 2018) with $R^2 = 0.9998$.